Intro
On Sep 29, Max Morawski opened an "Am I the Jerk (AITJ)" survey for the students in his Introduction to Data Science class. The survey also included questions about personal info, like school year and age, and habits, like church attendance. 169 submissions were recorded. UMD's database lists total enrollment in the class at 240 students, so the entirety of the class is not represented. This report explores correlations between responses to survey questions, especially between habits and AITJ judgments.

Background
The data from Max's survey included the variables:

- Timestamp
- School Year
- Age
- Parent and Personal Political Standing
- Parent and Personal Church Attendance Rate
- Parent and Personal Sports Viewership Rate
- Gender
- 14 AITJ questions (answers range from "Not a jerk"–"Strongly a jerk")

For cleaning purposes, the first response was dropped because it was missing responses. In order to better visualize the AITJ questions, the column names were changed to letters instead of the questions themselves. The timestamp data was converted from strings to floats/decimals, then to a "Time of day" variable that describes at what hour a response was submitted.

While exploring how men responded to male posters, responses to AITJ were mapped to numbers: (seen more in findings)

- 0: Not the jerk
- 1: Mildly the jerk
- 2: Strongly the jerk

This was done to calculate the mean response to each question, which needs numeric data.

AITJ questions had to be labeled "Male", "Female", and "Unknown" to make exploring responses to male questions possible. This was used to create a smaller dataset: a row of male averages to male questions and a row of female averages to male questions. I ran into an issue visualizing and testing this dataset because I wanted to graph and compare the rows, but scipy and matplotlib functions are usually done to the columns. This was simply fixed by transposing the dataset so that there existed a column of all male averages and a column of all female averages.

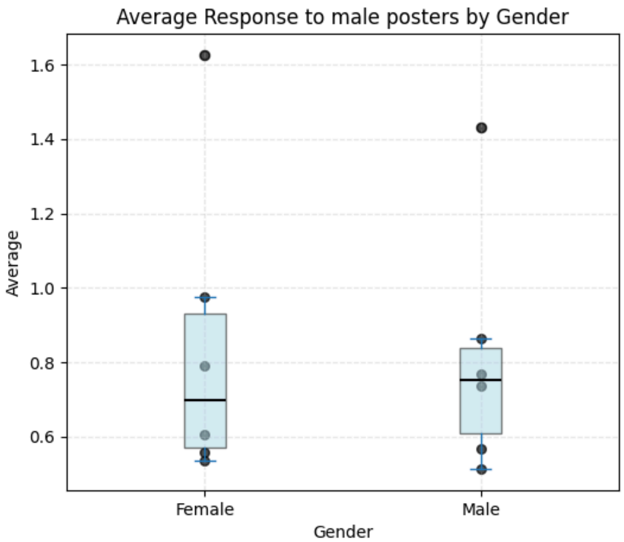
Lastly, the responses for Church Attendance rate were mapped:

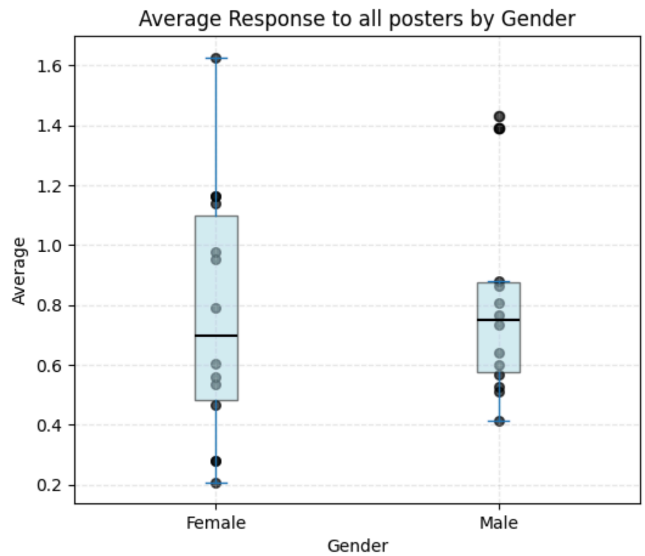
- "Frequently" and "Often": "Churchgoer"
- "Never": "Nongoer"

This is because there were very few Often and Frequently responses compared to Never.

Findings

Lack of gender bias

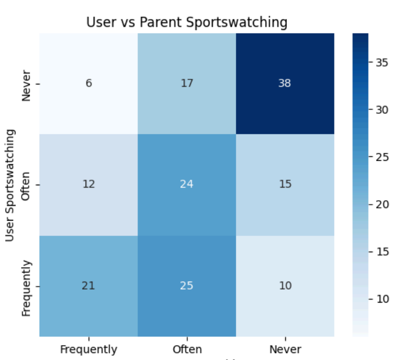


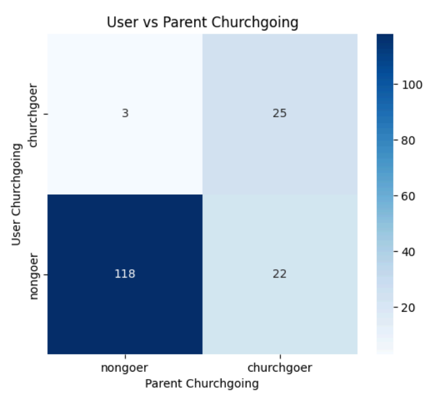


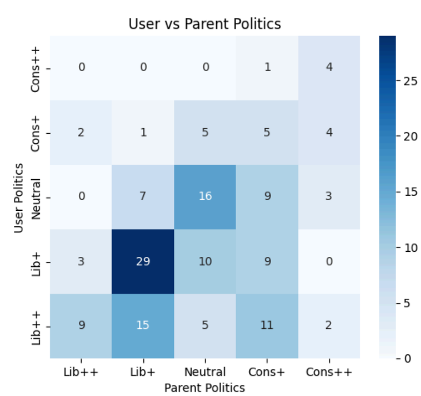
The average of female and male judgments were nearly identical–conveyed by the black line inside the 4 charted boxes–both when responding exclusively to male posters and when responding to all posters. A correlation between gender and average judgment could not be proven by a two-sample t-test, which returned a p-value of .874 for responses to male posters .699 for responses to all posters, well above the common alpha level of .05.

Link between parents' habits and respondents' habits

A strong correlation was seen between parent and respondent habits for political standing, sports viewership, and church attendance. The darkened upward diagonal on the graphs below suggests that parent and respondent habits typically match. The third graph also shows that the majority of respondents and their parents lean liberal, seen in the darkened left and lower parts of the graph.

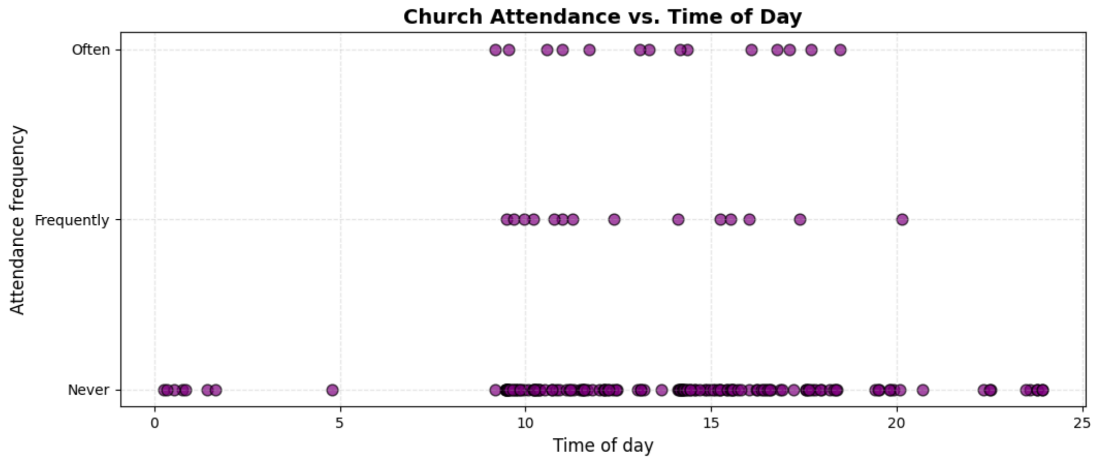




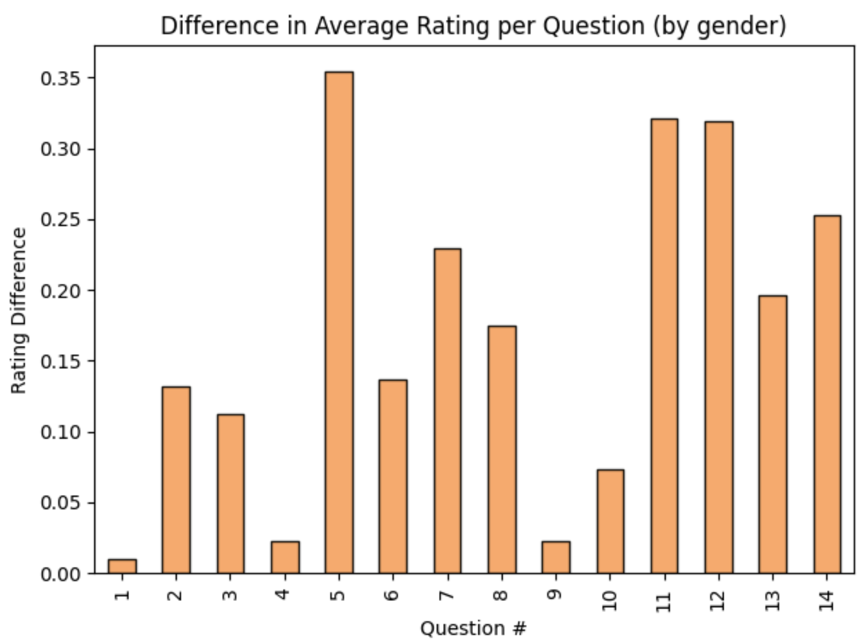


Other findings

All the respondents who answered in the middle of the night (12AM - 6AM) never attend church.



Comparing male responses to female responses, the most polarizing question–the question with the largest difference in average judgement–was question 5, which discussed paying for a wife's son to attend private school.



Conclusion

The analysis suggests that students largely follow many of the same traits as their parents, at least when it comes to sportswatching, churchgoing, and political standing. It could also be concluded that the majority of UMD students are moderately liberal with moderately liberal parents, but larger data with more variety is needed for that generalization. While these relationships were seen in the report, I believe it's fair to say these two conclusions were obvious and could have been determined without the data from this survey.

No correlation was seen between a respondent's gender and their average AITJ rating. It also couldn't be concluded that men vote differently from women when the AITJ poster is male. One interpretation is that gender bias does not have as strong an influence on judgment as we might have thought. I believe a more likely interpretation is that AITJ questions may not be the best medium to explore trends in gender, since responses are typically the same regardless of respondent background. A more suitable method might be questions about gender norms, since:

- there's less of a "correct" answer, like it feels like there is in AITJ questions
- respondents may feel they are better reflected in the questions
- Respondent gender and gender norms are closely related by nature

Lastly, further study could be done for when the AITJ poster is female, which wasn't possible with the current data because of an underrepresentation of female posters.